

Enterprise Manufacturing Business Intelligence System (EMBIS)

Technical Architecture

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Version: 1.0

Technology Stack: Microsoft Excel • Power Query • Power Pivot • DAX

1. Solution Overview

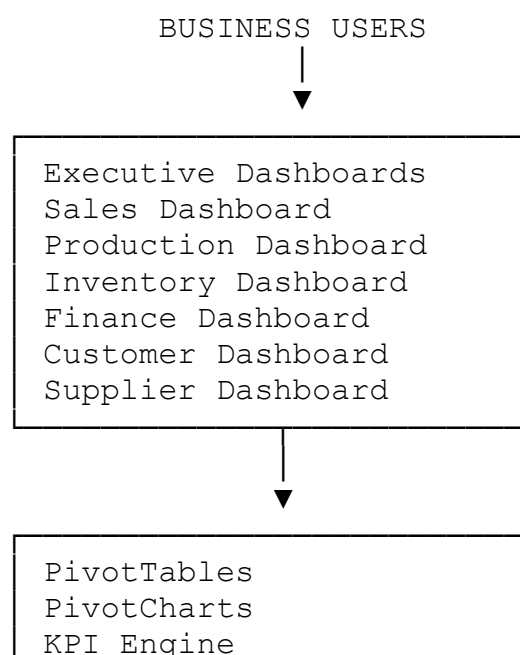
The Enterprise Manufacturing Business Intelligence System (EMBIS) is an end-to-end Business Intelligence solution developed using Microsoft Excel.

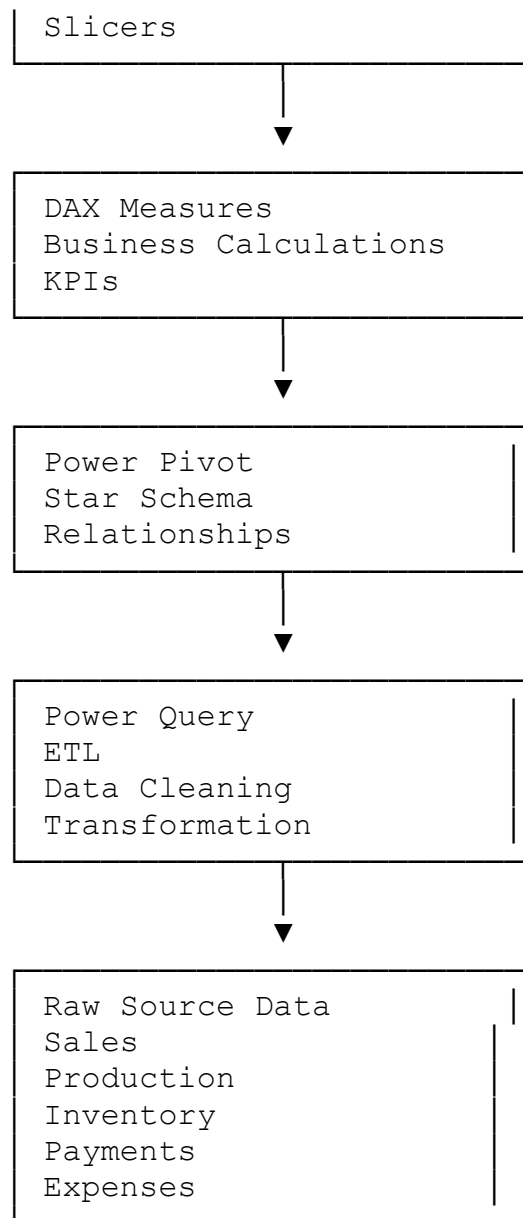
The solution follows a layered architecture that separates:

- Data Storage
- Data Preparation
- Data Modelling
- Business Logic
- Reporting
- User Interaction

This approach improves scalability, maintainability, and reporting performance.

2. Architecture Diagram

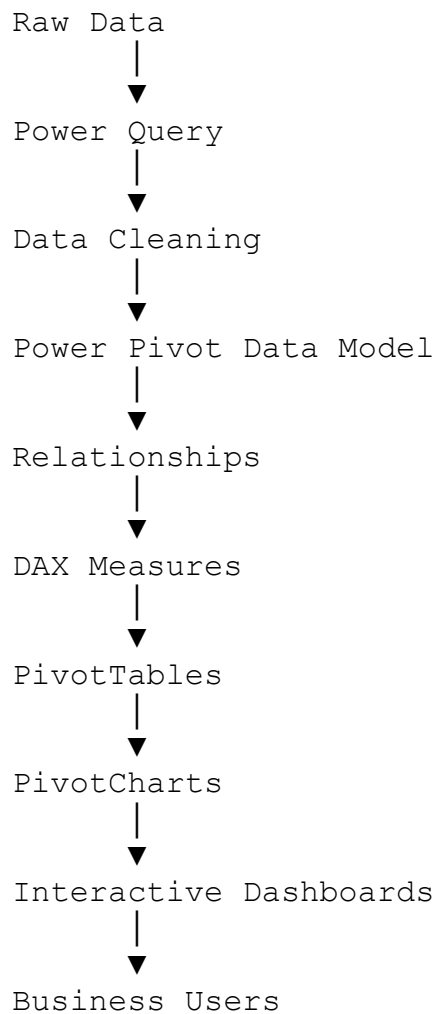




3. Solution Components

Layer	Description
Raw Data	Manufacturing transactional data
Power Query	Extract, Transform and Load (ETL)
Power Pivot	Data modelling and relationships
DAX	KPI calculations and business logic
Pivot Layer	PivotTables and PivotCharts
Dashboard Layer	Interactive executive dashboards

4. Data Flow



5. Dashboard Architecture

Dashboard Primary Business Function

Executive	Enterprise Performance
Sales	Revenue Analysis
Production	Manufacturing Performance
Inventory	Inventory Management
Finance	Financial Performance
Customer	Customer Analytics
Supplier	Supplier Performance

6. Key Technologies

Technology	Purpose
Microsoft Excel Reporting Platform	
Power Query	ETL
Power Pivot	Semantic Data Model

Technology	Purpose
DAX	KPI Calculations
PivotTables	Data Aggregation
PivotCharts	Data Visualization
Slicers	Interactive Filtering

7. Design Principles

The EMBIS solution was designed using the following Business Intelligence principles:

- Star Schema data modelling
 - Separation of ETL, modelling, and reporting layers
 - Reusable DAX measures
 - Consistent dashboard layouts
 - Interactive filtering using slicers
 - Modular dashboard architecture
 - Centralized KPI calculations
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8. Technical Highlights

- Automated ETL with Power Query
 - Optimized Star Schema data model
 - Five Fact Tables
 - Six Dimension Tables
 - Centralized DAX KPI Engine
 - Seven Interactive Dashboards
 - Standardized navigation system
 - Cross-dashboard slicers
 - Professional dashboard layout
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9. Future Architecture Enhancements

- Migration to Microsoft Power BI
 - SQL Server data warehouse
 - Automated scheduled refresh
 - Python ETL pipeline
 - Cloud deployment
 - Predictive analytics and forecasting
 - Row-level security
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Document Information

Item	Details
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